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Social CEOs: The effects of CEOs' communication styles and parasocial interaction on social networking sites

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Abstract
As one of the first empirical analyses focusing on corporate executive officers' (CEOs) communication style on social media and its impacts, the study aims to advance the theoretical understanding of CEO sociability and the mechanism underneath its effects on the broader public relations outcome of organization–public relationship and public advocacy. Through a quantitative survey with 332 social media users who have followed corporate CEOs on social media, this study illuminated how CEOs' responsive and assertive communications induce the CEOs' followers to perceive the corporate leaders as amicable role models and caring friends, which in turn improve their trust of, satisfaction with, and advocacy for the company. Hence, the findings highlight the pivotal role of CEOs as chief engagement officer in developing meaningful interpersonal interactions and relationships with today's digital savvy publics.

Keywords
CEO communication, organization–public relationships, parasocial interaction, public advocacy, social media, socio-communication style

Recognizing the unprecedented power of social technologies for organizations to enhance connectivity and collaboration with their stakeholders, Edelman (2008)

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conceptualizes a paradigm shift from public relations to public *engagement* to highlight the changing dynamics of dialogue and interaction between publics and organizations. Kang (2014) further argued that the success of organizations in today's new media landscape heavily relies on effective public engagement strategies for meaningful partnership.

Moreover, recognizing the potential of social media to humanize monolithic organizations (Vidgen et al., 2013), an increasing number of corporate leaders including Hewlett-Packard's Meg Whitman and Virgin Group's Richard Branson have embraced social media to boost their visibility and sociability, enjoying a celebrity status with huge followings. Industry reports suggested that >80% of corporate executive officers (CEOs) surveyed believe that social media is an important platform for them to deepen connections with customers, employees, and investors (Brandfog, 2013). The interactive, personal, and relational features of social media help to reduce the power distance between CEOs and the publics and transform the distant, faceless images of CEOs by allowing them to communicate with their stakeholders in a personable manner. The symmetrical communication enabled on social media may further enhance a company's transparent communication represented by the accessibility and sociability of its CEO (Men & Stacks 2014). CEOs' social media presence and activeness are also believed to showcase innovativeness and enhance their companies' image and reputation (Weber Shandwick, 2012). Consequently, "social CEOs" who can engage their stakeholders in genuine conversations are believed to be the new wave of corporate leaders (Weber Shandwick, 2012).

In public relations, the CEOs' critical role has been widely acknowledged. Scholars argue that CEOs personify the organization as the highest level corporate spokesperson to communicate an organization's vision (Park and Berger, 2004; Ranft et al., 2006). A successful CEO not only enhances publics' identification with the organization and creates a favorable corporate reputation but also forges positive relations with various stakeholders that contribute to the company's bottom line (Ranft et al., 2006).

With rising demand of corporate transparency and authenticity driven by social media (Author, 2014), publics have come to expect greater access to the insights and vision of corporate leaders in more open dialogues. The image of CEOs has evolved to become more visible, social, and approachable than ever before (Brandfog, 2013). The 2014 Edelman Trust Barometer coined the term "Chief Engagement Officer" to highlight the pressing need for CEOs to step forward to engage with their stakeholders. Particularly on social networking sites (SNSs), CEOs' personal pages serve as a more personal, intimate venue that allows them to engage individual stakeholders and fans in interpersonal communications and, in turn, foster authentic leadership (Author, 2014). Thus, CEOs' social communications may work to personify and humanize their corporations and ultimately help their organizations to get closer to their online stakeholders (Vidgen et al., 2013). However, despite the increasingly influential role of CEOs in public relations and corporate communication, theoretical deliberations on how and why CEOs' social communication matters, as well as strategic guidelines on how CEOs should communicate in the social platforms, are seriously lacking. Focusing on leadership communication, this study examined the socio-communicative style of CEOs' SNS communications perceived by their followers and, more importantly, the effects of

CEOs' social communications on publics' interactions with the corporate leaders and their overall relationship with and support for the organization.

The potential of social media for strengthening organization–public relationships (OPRs) lies in its power of enabling and encouraging interpersonal communication and interaction. The emerging literature on interpersonal communication strategy on social media highlights its significance for public relations (Sung and Kim, 2014; Wasike, 2013). For instance, interpersonal messages characterized by being highly interactive and non-promotional were found to improve organizational evaluation (Sung and Kim, 2014). Building an engaging corporate character for boosting interpersonal communication via corporate SNS pages also effectively enhances favorable organizational reputation and meaningful relationships with stakeholders (Men & Tsai, 2015; Verčič and Verčič, 2007).

This study presents one of the first empirical investigations to (a) evaluate how CEOs' socio-communicative styles may stimulate interactions in the mediated social environment and (b) explore how CEOs' social communications with stakeholders on SNSs may lead to positive relational and behavioral outcomes of OPR and public advocacy. The study findings aim to advance the emerging literature on leadership communication and relationship management in the social media context. This study will also provide strategic guidelines for organizational leaders and public relations professionals to craft effective social communication strategies to boost personal, intimate interactions with online publics and, in turn, build long-term OPRs.

Theoretical framework

As top leaders of their organizations, CEOs influence organizational direction, relations with stakeholders, corporate reputation, and organizational effectiveness (Men, 2014; Resick et al., 2009). Particularly in news media, the CEO is often sought out as the highest authority to provide information on the organization (Kantola, 2014). Not surprisingly, media have been interested in the personality of the CEO, and corporate executives are constantly "pressed by shareholders and the public for reams of detailed information about their lives and activities" (Kantola, 2014: 31). Additionally, publics have come to expect and demand more direct and open interactions with CEOs in order to be part of the decision-making process. For instance, publics actively reached out and communicated with the CEOs of Chiquita Brands to influence the company's decision of relocating its corporate headquarter (Jameson, 2014). In this way, the publics' anticipated accountability of CEOs to respond to their opinions is likely to lead to a more ongoing, substantive relationship.

Regarding the study focus of leadership communication, a number of researchers have examined the role of CEOs as corporate spokespersons (e.g. Hutton et al., 2001; Park and Berger, 2004). To many stakeholders, CEOs represent the face and soul of the organizations (Garten, 2001). However, in sharp contrast to the excitement and speculations about the rise of social CEOs, little empirical research has examined CEOs' social communications and the associated effects. It is thus imperative to assess how CEOs' social communications may work to enhance interactions with stakeholders that lead to meaningful relationship and desirable public relations outcomes.

CEOs' socio-communicative style

To understand how CEOs should communicate via social media, this study focuses on the influence of CEOs' socio-communicative styles. According to Thomas et al. (1994), socio-communicative style refers to the different skills an individual uses in initiating, adapting, and responding to interpersonal communication. Analyzing the socio-communicative style provides "insight into the personality of individuals by taking note of their characteristic communication behaviors" (Thomas et al., 1994: 109).

Specifically, socio-communicative style consists of two dimensions—assertiveness and responsiveness—that are conceptualized to represent the masculine and feminine qualities of communication, respectively (Thomas et al., 1994). Assertiveness refers to one's ability to be independent, dominant, and to stand up for oneself, whether to make a request, defend one's belief, or express a feeling (McCroskey and Richmond, 1996). An assertive communicator is described as dominant, forceful, and competitive. In comparison, responsive communicators tend to be sensitive and empathic to others. Thus, responsive communicators are often described as friendly, empathetic, vicarious, understanding, and ultimately a good listener (McCroskey and Richmond, 1996).

The effects of assertive and responsive socio-communicative style have been examined predominantly in the education setting. For instance, teachers' assertive and responsive communications are positively related to students' trust (Wooten and McCroskey, 1996). Assertive teachers are believed to be more competent, less likely to exhibit classroom misbehaviors, and improve student affect toward the teacher (Henning, 2010). Additionally, teacher responsiveness is associated with students' liking for the teacher and even course content. Importantly, McCroskey and Richmond (1996) point out that each style has value, and no one style is necessarily better than the other. Considering the CEOs' top leader position and the relationship orientation of social media, CEOs' social communications are expected to demonstrate a mix of assertive and responsive styles. In other words, a CEO may be judged by the publics as more competent and reliable if he or she is able to logically and confidently answer questions, provide advice, or defend beliefs in an assertive manner. At the same time, CEOs' responsiveness of being compassionate and listening to publics' concerns may also improve their interactions with the publics.

Parasocial interaction

To illuminate the power of social media as a relationship management tool, we focus on how CEOs' social communications influence stakeholders' parasocial interaction (PSI) with them. Defined as audiences' perceived intimate and personal relationship with mediated characters whom they viewed as guides, friends, and companions (Horton and Wohl, 1956; Rubin, 2002), PSI has been extensively studied in the television and entertainment literature to understand audience's relationship with media personalities. Through repeated, frequent, and prolonged media consumption, viewers become familiar with the media characters' opinions, values, mannerism, sense of humor, and other personality characteristics (Eyal and Dailey, 2012; Stever, 2013). They thus develop feelings of being personally connected to the mediated characters.

The construct of PSI can be particularly instructive for the understanding of mediated interactions on social media. Specifically, PSI provides a common ground for mass media and interpersonal communication (Rubin and McHugh, 1987) and echoes the unique characteristic of social media that blurs the boundary between mass and personal media. Moreover, the interactive environment of social media makes PSIs in such context potentially more tangible. Most PSIs involve a status difference, "with the parasocial object being a person of higher status, most often through wealth, fame or power, than the viewer" (Stever, 2013: 4). However, the personal nature of social media serves to humanize the CEOs, bringing them closer to the publics, and thus may engender stronger PSIs.

Scholars have conceptualized PSI as a multidimensional construct that involves viewers' cognitive, affective, and behavioral engagement with media personalities (Papa et al., 2000). For instance, Tsay and Bodine's (2012) conceptualization of PSI involves such aspects as guidance (e.g. learning from a media personality as a role model), desire for face-to-face contact (e.g. wanting to meet the media personality), intimacy (e.g. perceiving closeness outside of media consumption setting), and behavioral engagement (e.g. talking to others about the media personality).

Key characteristics of media personalities that tend to inspire the formation of PSI have also been identified, including attractiveness, predictability, and notably, conversational speech (Houlberg, 1984; Rubin et al., 1985). In television consumption, viewers' parasocial processing is intensified when a persona addresses users directly (Schramm and Hartmann, 2008), which echoes the interpersonal approach in corporate communications. Recent studies on social engagement underline the interpersonal approach to resemble interpersonal conversations in dialogic, human voice to effectively engage stakeholders (Sweetser and Metzgar, 2007). For instance, corporate communicators often act as "friends" to better blend into their stakeholders' social networks (Men & Tsai, 2012). Different from organizations' SNS communicators who act as anonymous corporate representatives, CEOs could be well-known public figures, and their social communications may entail higher authenticity and personality that further boost PSI (Rosaen and Dibble, 2008; Tsay-Vogel and Schwartz, 2014). On SNSs, stakeholders can directly observe how the CEO interacts with the publics on various agenda via wall posts and get to know the CEO as an identifiable and personable "friend." At the same time, the CEOs remain a mediated persona, and their communications and connections with stakeholders are transmitted through social media (Jiang et al., 2010). Following Thorson and Rodgers' (2006) approach, this study updates the definition of PSI that has been limited to traditional media by redefining PSI as a user's interpersonal involvement with a media personality (including CEOs on social media) through mediated communication.

Prior studies exploring users' PSI with public figures on social media demonstrate that such interactions are serious, meaningful, and impactful for the users (Stever and Lawson, 2013). PSI was found to mediate politicians' web communication as a form of electronic word of mouth (eWOM; Thorson and Rodgers, 2006). Similarly, Colliander and Dahlén (2011) found that the effectiveness of brand publicity on blogs on consumer response is mediated by PSI. Focusing on politicians' Twitter communications, Lee and Shin (2014) reported that although the contents were identical, compared to traditional

media, the politician's messages on Twitter heightened a sense of direct conversation (i.e. social presence), which, in turn, induced more favorable impressions of and a stronger support for the politician. Indirect support based on CEO blogs also suggests that the ability of CEOs to engage users on blogs is essential to achieve community building and genuine dialogues (Vidgen et al., 2013). Thus, the following hypotheses are formulated to test the effects of CEOs' responsive and assertive communications on SNSs to boost PSI with stakeholders:

H1. CEOs' responsive communication positively influences PSI.
H2. CEOs' assertive communication positively influences PSI.

OPRs

The proliferation of social technologies that enable publics to engage one another via computer-mediated communication is significantly transforming relational dynamics (Pauley and Emmers-Sommer, 2007). On social media, CEOs' efforts to communicate their personal beliefs and values and respond to stakeholders' concerns may encourage the development of sustainable, satisfactory relationships with the publics and, by extension, with the organizations they lead and represent. However, while there has been a rapid growth of studies on how social media affect and alter interpersonal relationships (e.g. Pauley and Emmers-Sommer, 2007; Peter and Valkenburg, 2007), there exist little empirical data to address how CEOs' communications and interactions with stakeholders on social media affect stakeholders' relationship with organizations.

Despite the dearth of empirical evidence on the effects of socio-communicative styles on relationships in the context of corporate communication, research focusing interpersonal communication suggests the significant potential. Because relationships are created and maintained through communication, the extent to which an individual is perceived as assertive or responsive may affect others' perceptions and hence the extent to which others are willing to enter into a relationship with the individual (DiClemente et al., 2013). More importantly, Sung and Kim (2014) reported that organizations' interpersonal communication strategies featuring non-promotional, personal, and dialogic messages on corporate social accounts contribute to the formation of OPRs. Along this line of reasoning, CEOs' perceived responsive communication style on SNSs will have a positive impact on OPRs.

With regard to the assertive dimension of communication styles, research shows that assertive communication increases perceived interpersonal trust, which is one of the most essential factors in relationships (Snavely, 1981). Additionally, studies on the relevant construct of self-assertion that refers to the capability to express oneself to others without interpersonal anxiety demonstrate that self-assertion on SNSs is positively associated with the strength of both bonding and bridging social relationships (Jin, 2015). Assertiveness is also positively related to communicative competence (Zakahi, 1985), which in turn affect relationship satisfaction and commitment (Arroyo and Segrin, 2011). The following hypotheses are provided to empirically test the connections between assertive and responsive communication styles and OPRs:

H3. CEOs' responsive communication on social media positively influences OPRs.
H4. CEOs' assertive communication on social media positively influences OPRs.

To further gauge the enduring effects of PSI, scholars have theorized viewers' long-term parasocial relationship (PSR) with media persona constructed through various mediated encounters (Claessens and Van den Bulck, 2015). The effects of PSI for audience enjoyment, learning, identification, and need gratification in media consumption have also been widely documented (Tsay and Bodine, 2012). However, prior studies are limited to TV celebrities in entertainment and fandom studies (e.g. Bocarnea and Brown, 2007; Hartmann et al., 2008). While research suggests that a strong PSI can enhance positive attitudes and intentions toward the media programs (Oliver, 2009; Tsay and Bodine, 2012), the broader impact of PSI on relationship beyond the interpersonal context has not been sufficiently analyzed. To explore the potential of PSI as a public relations tool for relationship management, this study proposes that publics' mediated interactions with CEOs on social media will affect their relationships with the organizations led by the corporate executives:

H5. PSI of the CEO positively influences OPRs.

Public advocacy

In order to enrich the theoretical knowledge of CEO sociability, this study went a step beyond OPRs and explored the important behavioral outcome of public advocacy, which has been recognized as a critical effect of the strong bond between an organization and its publics (Walz and Celuch, 2010). According to Walz and Celuch (2010), public advocacy refers to publics' behaviors of actively and voluntarily promoting a company through positive word of mouth or defending against critics for the company, its products, or its brands. Due to the communal, participatory, and collaborative orientation of social media communications, publics are naturally and astutely aware of the powerful impact of their advocacy or criticism among their social networks. Such consciousness and the associated sense of empowerment can then increase the likelihood that they will engage in advocacy activities (Pınar Özdemir, 2012).

Recent studies have explored antecedents of public advocacy, including corporate trust, source credibility, and quality of corporate communication (Krapfel, 1985; Walz and Celuch, 2010). Urban (2005) argued that a company must provide open, honest, and complete information to encourage consumer advocacy, suggesting the importance of CEO to engage stakeholders in meaningful conversations and interactions. Therefore, public advocacy was examined as a crucial outcome of PSI and OPRs in this study.

Research shows that PSI affects how mediated messages are interpreted and even dictates viewers' knowledge, attitudes, and behaviors in the real world (Brown and Basil, 1995; Moyer-Guse, 2008). Kassing and Sanderson (2010) suggested that the interactive and communicative features of social media stimulate the behavioral outcome associated with fans' PSI with the celebrities. Sanderson (2008) further argued that fans often actively display their PSR and its significance to others. Also reporting the behavioral effect, Sood and Rogers (2000) argued that that PSI can evoke interpersonal reactions, behavior change,

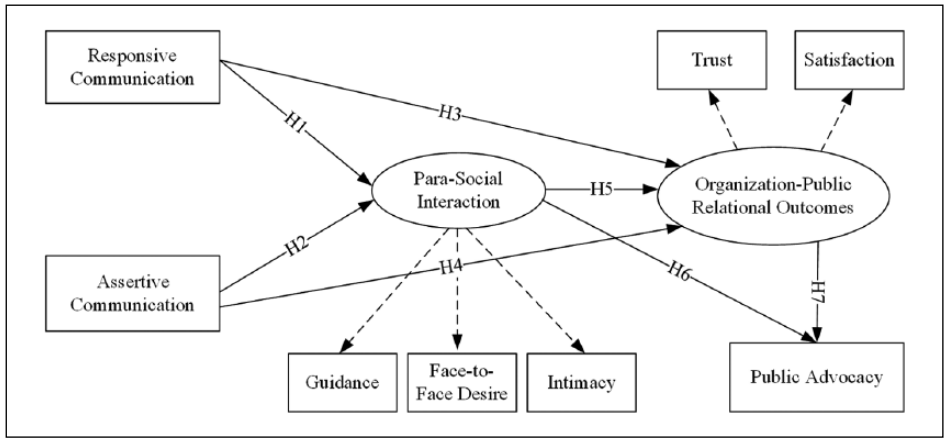


Figure 1. Proposed model of the effects of CEOs' communication style on social media and parasocial interaction on organization–public relationship outcomes and public advocacy.

and external dialogue. Focusing on blogs, Thorson and Rodgers (2006) similarly argued that messages from media sources where perceived PSI is high can lead to positive effects similar to those of eWOM communication. Therefore, it is expected that stakeholders' PSI with a CEO on SNSs may translate into advocacy behaviors for the CEO's company:

H6. PSI positively influences public advocacy.
H7. The quality of OPRs positively influences public advocacy.

Methodology

To test the proposed model presented in Figure 1, we conducted an online survey using random sample drawn from online survey panels managed by an international research firm in July 2014. The population of the study comprised adult American users of SNSs. Only SNS users who had followed or liked at least one CEO of a company on SNSs (e.g. Facebook, Twitter, and LinkedIn) were included in this study. The final sample included 332 SNS users from diverse demographic backgrounds. To achieve a representative sample, we applied stratified and quota random sampling strategies to solicit respondents from different age groups, genders, income, and education levels. The average age of the respondents surveyed was 36 years. Among the respondents, 59% were women and 41% were men, and 74% of the respondents had received some college education and above. In terms of SNS usage, approximately 86% of the respondents had used SNSs for >3 years, and around half of the respondents (47%) spend 1–2 hours/day on SNSs, and 30% respondents spend at least 3 hours/day on SNSs. Around 61% respondents follow one or two CEOs on SNSs, 30% follow three or four CEOs on SNSs, and 10% of the respondents follow over five CEOs on SNSs. In the beginning of the survey, respondents were first instructed to identify and review a CEO's SNS page they had recently visited (e.g. browse, comment, or post) and list the

name of the CEO's company. They then answered questions regarding the CEO's communication style perceived via his or her SNS posts and the respondent's PSI with the CEO as well as relationship with the company.

Measures

All measurement items used in this study were adopted from previous literature and adapted to fit this study context (see Appendix 1). A 7-point Likert scale was used to measure the focal constructs in this study. An online pretest was first conducted with 80 randomly selected respondents from the research panel maintained by the same research firm. Preliminary analysis of the reliability and validity of the measures demonstrated satisfactory results.

The Assertiveness–Responsiveness Measure (Richmond and McCroskey, 1990), which includes nine items related to communication responsiveness (e.g. "sincere," "friendly," and "compassionate," $\alpha = .96$) and six items related to assertiveness (e.g. "act as a leader," "dominant," "aggressive," "competitive," and "has a strong personality," $\alpha = .85$), was used to evaluate CEO's communication style. A total of 17 items adapted from Tsay and Bodine's (2012) and Eyal and Te'eni-Harari's (2013) studies were used to measure the multidimensional PSI, namely, the aspects of guidance, face-to-face desire, and intimacy between the CEO and their followers. Specifically, guidance was measured with seven items (e.g. "I treat the CEO as a role model" and "I use advice that I learn from the CEO on his or her SNS pages," $\alpha = .94$). Face-to-face desire was measured with four items (e.g. "I would be happy to meet the CEO in person" and "I would be comfortable with the CEO if we met in person," $\alpha = .88$), and intimacy was assessed with six items (e.g. "I miss the CEO of the company when there are no updated SNS wall posts" and "I think the CEO can be a good friend of mine," $\alpha = .93$). Hon and Grunig's (1999) scale was adopted to measure organization–public relational trust ($\alpha = .92$; "I feel very confident about this company's skills") and satisfaction ($\alpha = .93$; "I am happy with this company."), the two relational outcomes most relevant to the focal concepts of this study. Finally, five items adapted from Author (2014) were used to assess public advocacy toward the company that the CEO represents (e.g. "I say positive things about this company to people I know" and "I defend this company when someone says something negative about it," $\alpha = .93$).

For data analysis, descriptive analysis and structural equation modeling (SEM) were conducted. Multiple criteria were used to evaluate the goodness-of-model fit, including the comparative fit index (CFI), the root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR) indices, which are a minimal set of fit indexes that should be reported and interpreted when reporting the results of SEM analyses (Kline, 2005).[1]

Results

Preliminary analysis of the data showed that respondents were most likely to follow CEOs on Facebook ($n = 232$, 69.9%), followed by Twitter ($n = 63$, 19.0%) and LinkedIn ($n = 24$, 7.2%). Other SNS platforms reported for following CEOs included YouTube ($n = 4$, 1.2%), Instagram ($n = 4$, 1.2%), Tumblr ($n = 3$, 0.9%), and Google+ ($n = 2$, 0.6%). About 69% of

Table 1. Descriptive of CEOs' social media communication style, parasocial interaction, organization–public relational outcomes, and public advocacy (*M, SD*, and correlations).

	α	*M*	*SD*	1	2	3	4	5	6	7	8
1. Responsiveness	.96	5.27	1.74	1.00							
2. Assertiveness	.85	5.13	1.08	.36**	1.00						
3. Guidance	.94	4.97	1.27	.71**	.38**	1.00					
4. Face-to-face desire	.88	5.43	1.17	.59**	.35**	.65**	1.00				
5. Intimacy	.93	4.52	1.36	.66**	.27**	.79**	.69**	1.00			
6. Trust	.92	5.20	1.17	.76**	.29**	.67**	.56**	.64**	1.00		
7. Satisfaction	.93	5.36	1.17	.68**	.28**	.60**	.50**	.55**	.84**	1.00	
8. Public advocacy	.93	5.08	1.27	.59**	.28**	.63**	.52*	.63**	.69**	70**	1.00

CEO: corporate executive officers; *SD*: standard deviation.
**Correlation is significant at $p < .01$ (two-tailed).

the CEOs whom the respondents followed were founders of the companies. The companies of the CEOs that respondents followed ranged from well-known global corporation to small, local, new businesses and across various industries. The majority (70%) of respondents had followed the CEO for <2 years. As shown in Table 1, CEOs' communications on social media were perceived to be a mix of responsive ($M = 5.27$, standard deviation [*SD*] = 1.74) and assertive styles ($M = 5.13$, $SD = 1.08$). In terms of PSI, respondents on average reported higher on face-to-face desire ($M = 5.43$, $SD = 1.17$) and guidance ($M = 4.97$, $SD = 1.27$) than on perceived relational intimacy ($M = 4.52$, $SD = 1.36$). Overall, respondents exhibited trust ($M = 5.20$, $SD = 1.17$), satisfaction ($M = 5.36$, $SD = 1.17$), and advocacy ($M = 5.08$, $SD = 1.27$) toward the company whose CEO they followed on social media.

The analysis of the conceptual model involved a two-stage process: (a) an assessment of the construct validity of the measurement model using confirmatory factor analysis and (b) an assessment of the structural model. OPRs and PSI in the structural model were considered as latent variables. Responsive and assertive communications and public advocacy were treated as observed variables. The maximum likelihood method was employed for model estimation. The test of the measurement model demonstrated satisfactory fit to the data: $\chi^2(13) = 40.90$, $p < .001$, $\chi^2/df = 3.15$, RMSEA = .08 (90% confidence interval: [.05, .10]), SRMR = .02, Tucker–Lewis index (TLI) = .97, and CFI = .98. The standardized factor loadings between latent variables and observed variables ranged from .75 to .95, indicating satisfactory construct validity. Cronbach's alphas for composite construct variables of OPRs and PSI were .95 and .96, respectively. Cronbach's alphas for observed variables in the model ranged from .85 to .96 as presented in Table 1.The hypothesized structural model displayed in demonstrated satisfactory fit to the data: $\chi^2(15) = 46.69$, $p < .001$, $\chi^2/df = 3.11$, RMSEA = .08 (90% confidence interval: [.05, .10]), SRMR = .02, TLI = .97, and CFI = .98.[2] Five structural paths demonstrated significant results.

Hypothesis testing

The study results supported H1 and H2 that predicted positive effects of CEOs' responsive and assertive communications on PSI (see Figure 2). In particular, CEOs' responsive

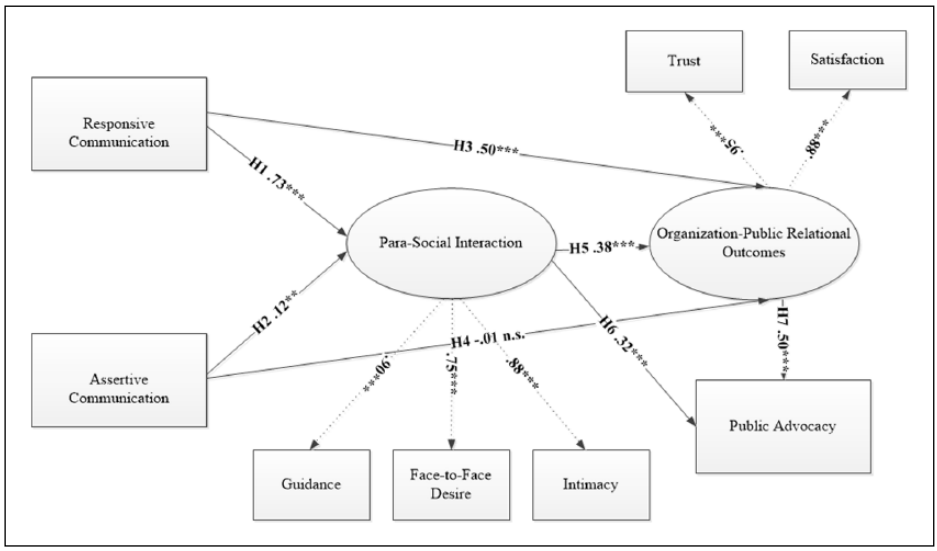


Figure 2. Results of the hypothesized model. Coefficients are standardized regression weights. For the sake of brevity, error terms of indicators and disturbances of endogenous variables were omitted from this figure.
***$p < .001$, **$p < .01$.

communication demonstrated a strong positive effect on PSI ($\beta = .73$, $p < .001$), whereas CEOs' assertive communication exhibited significant but weak effect ($\beta = .12$, $p < .01$). Therefore, although both communicative responsiveness and assertiveness help CEOs to form interpersonal relationship with online publics, the responsive communication style had a particularly decisive influence.

H3 and H4 predicted the positive effects of CEO's communication style on the quality of OPRs. CEO's responsive communication was found to demonstrate a large positive effect on OPRs ($\beta = .50$, $p < .001$). In other words, when a CEO's social communication is perceived as responsive, sincere, and compassionate, publics tend to express a higher level of trust and satisfaction with the company. However, contrary to the prediction, CEOs' assertive communication showed non-significant effect on public trust or satisfaction, thus rejecting H4.

Additionally, H5 and H6 that proposed the positive impact of PSI on OPRs and public advocacy were both supported by the results. PSI revealed a large positive effect on the quality of OPRs ($\beta = .38$, $p < .001$) and public advocacy ($\beta = .32$, $p < .001$). Therefore, publics' interactions with CEOs mediated via SNSs were found to have a broader, favorable impact beyond the interpersonal context and contributed to stakeholders' trust and satisfaction with the company, and ultimately their advocacy for the company. Finally, publics' H7 that predicted the positive effect of OPRs on public advocacy was also supported ($\beta = .50$, $p < .001$), indicating that publics who possess better relationships with the company are more likely to advocate for the company publicly or defend the company in times of turbulences or crises.

Discussion and conclusion

As one of the first empirical analyses focusing on CEOs' communication style on social media and its impacts, the study findings advance the theoretical understanding of CEO sociability and the mechanism underneath its effects on the broader public relations outcome of OPRs and public advocacy. Specifically, this study illuminated how CEOs' responsive and assertive communications induce the CEOs' followers to perceive the corporate leaders as amicable role models and caring friends, which in turn improve their trust of, satisfaction with, and advocacy for the company. Given the unprecedented public demand for transparency and openness in today's business world (Author, 2015), corporate leaders' social communication that is responsive, empathetic, accountable, and decisive is key to cultivating the company's symmetrical and transparent communication system (Author, 2014). Hence, the findings highlight the pivotal role of CEOs as chief engagement officer in developing meaningful interpersonal interactions and relationships with today's media savvy publics.

Theoretically, by demonstrating the effects of CEOs' socio-communicative style on public relational and behavioral outcomes toward the organization, the study expands the growing knowledge on leadership communication and proves the notion that strategic leadership and sociability matters for public relations success (Author, 2014; Grunig et al., 2002). It also extends the application of the socio-communicative style to the leadership communication and social media contexts. CEOs' responsive communication that positions them as empathetic, caring friends whom publics aspire to meet in person has a notably prominent role in strengthening their followers' belief in the relationship they have with the CEOs. In a way, the responsive communicative style with personal messages and a down-to-earth tone is more fitting with the social and personal environment of SNSs. At the same time, CEOs' assertive communication reflecting their opinion leader position also contributes to publics' PSI motivated by guidance-seeking. Such findings thus underscore the effectiveness of the interpersonal approach for leadership communication on social media where publics expect to get to know the corporate executives at a more personal level. Moreover, while prior studies documented the effect of PSI on audiences' enjoyment, identification, and learning in the media consumption context, this study provided empirical evidence for the broader impacts of PSI on OPR management (Author, 2015; Toth, 2000). The observed positive impact of relational trust and satisfaction on public advocacy behavior further testifies the value of quality OPRs for the organization.

Strategic implications

The study findings also provide important implications for communication professionals and top management on how to enact the role of CEOs as chief engagement officers and how to use social media to cultivate OPRs and encourage public advocacy (e.g. Sung and Kim, 2014; Sweetser and Metzgar, 2007). In general, although CEOs as top leaders are expected to speak up and be assertive and even dominant, the responsive communication style should be adopted for relationship-building purposes. Especially on social media where the communication hierarchies have been reduced, CEOs' images may be expected

to transform from being distant and intimidating to being approachable and amicable. Blending a professional and personal tone, CEOs could come into life, communicate as a friend, and actively foster conversations and personal relationships with their social media followers. CEOs should actively respond to followers' posts and communicate in a responsive, friendly, and empathetic manner to demonstrate their concern for various stakeholders and willingness to listen. They should also participate in community discussions and encourage interactions among members of the SNS community to build genuine bond with publics. Such responsive communication will reward the corporate leaders and their organizations with meaningful interactions and relationships with the publics.

Directions for future research

Despite the original findings and important implications, this study involves several limitations. Specifically regarding CEOs' new role as chief engagement officers, this study did not directly assess publics' engagement level and activities on CEOs' SNS pages which should be addressed in future research. Additionally, this study explored CEOs' assertive and responsive communication styles as perceived by the publics. However, it remains unclear what messages are perceived as assertive or responsive on social media and, in particular, how assertive style can be effectively applied for relationship-oriented and customer-focused messages. Content analyses should be conducted to review top-rated social CEOs' messages to achieve an in-depth understanding of the nature and expression of different socio-communicative styles in the social media setting. Although assertive behaviors have been stereotypically associated with masculinity and responsive behaviors have been equated with femininity (Mottet and Beebe, 2006), there exist mixed research findings regarding the connection between gender and socio-communicative styles (Anderson et al., 1997; Martin and Anderson, 1996; Thompson and Klopf, 1991). Future research should pay special attention to the effect of CEOs' gender on the perceived socio-communicative styles of their social media communications. In a related vein, it is likely that assertive and responsive communicative styles are perceived differently across internal and external stakeholder groups. Future research should assess CEOs' assertive and responsive messages via qualitative focus groups with various stakeholder communities.

In regard to PSIs between CEOs and stakeholders via social media, case study and qualitative interviewing with SNS users can be instructive to provide an in-depth exploration of netizens' PSI and relationship with corporate leaders. In general, PSI has not been sufficiently addressed in the settings of corporate communication and leadership communication. How PSI may affect other important perceptual outcomes such as perceived organizational reputation, corporate character, and image remains unexplored, and further empirical investigations are thus called for. Furthermore, the issue of CEO's communication credibility on social media is worth further deliberation. Gallicano et al.'s (2013) research on ghost blogging suggests that many organizational blogs reportedly authored by executives are often in fact written by someone else, such as their public relations agents. Ghostwriting likely occurs in other social media platforms, such as CEO's SNS pages, considering the amount of time required for personally managing the proliferated social media platforms (Briones et al., 2011). Thus, future study should

explore online publics' expectations and perceptions of CEO's social media communications in relation to ghostwriting and the associated effects on message credibility, PSI as well as perceived corporate transparency, authenticity, and trust. Other effective CEO's communication strategies identified in the literature should also be evaluated to allow a holistic understanding of the emerging phenomenon of social CEOs. By the same token, future endeavors should connect CEOs' social communication with other mediators (e.g. level of engagement and empowerment) and outcomes (e.g. loyalty, purchase and investment behavior, and business bottom line). Finally, the cross-sectional survey design is limited in suggesting the order of effects. Despite the application of SEM to remedy the issue, to establish the true causal links between the constructs, longitudinal studies or experimental approaches should be utilized in future research.

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Notes

1. According to Hu and Bentler (1999), a cutoff value close to .95 for comparative fit index (CFI) and Tucker–Lewis index (TLI), a cutoff value close to .08 for standardized root mean square residual (SRMR), and a cutoff value close to .06 for root mean square error of approximation (RMSEA) indicate good fit between the hypothesized model and the observed data. Additionally, according to Hu and Bentler's (1999) joint cutoff criteria, a structural equation modeling (SEM) model with CFI and TLI $\geqslant .95$ and SRMR $< .10$ or RMSEA $\leqslant .06$ and SRMR $\leqslant .10$ suggests that the fit between the data and the proposed model is reasonable.
2. The model was estimated by allowing the disturbances of perceived corporate executive officer's (CEO) authenticity and approachability to covariate because the two variables share a common factor perception of the CEO.

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Appendix 1

Corporate executive officer's communication style

Responsiveness

1. A good listener;
2. Sensitive to the needs of publics;
3. Interested;
4. Responsive;
5. Sincere;
6. Warm;
7. Friendly;
8. Compassionate;
9. Understanding.

Assertiveness

1. Has strong personality;
2. Dominant;
3. Forceful;
4. Aggressive;
5. Competitive;
6. Acts as a leader.

Parasocial interaction

Guidance

1. I treat the corporate executive officer (CEO) as a role model.
2. I am comfortable learning from the CEO.
3. I use advice that I learn from the CEO on his or her SNS pages.
4. I look up to the CEO.
5. The CEO teaches me important lessons.
6. I seek guidance from the CEO.
7. I often compare my ideas with what the CEO says on his or her SNS pages.

Face-to-face desire

8. I would be happy to meet the CEO in person.
9. If I saw the CEO on streets, I would talk to him or her.
10. I would be comfortable with the CEO if we met in person.
11. If given the opportunity, I would contact the CEO.

Intimacy

12. I miss the CEO of the company when there are no updated SNS wall posts.
13. I think the CEO could be a good friend of mine.
14. The CEO makes me feel comfortable, as if I am with a friend.
15. I think the CEO is like an old friend.
16. I see the CEO as a close friend.
17. I would like to have a friendly chat with the CEO.

Organization–public relational outcomes

Trust

1. Whenever the company makes an important decision, I know it will be concerned about people like me.
2. The company can be relied on to keep its promises.
3. I believe that the company takes the opinions of people like me into account when making decisions.
4. I feel very confident about the company's skills.
5. The company has the ability to accomplish what it says it will do.

Satisfaction

6. I am happy with the company.
7. Most people like me are happy in their interactions with the company.
8. In general, I am pleased with the relationship this company has established with me.
9. I enjoy dealing with this company.

Public advocacy

1. I say positive things about this company to people I know.
2. I defend this company when someone says something negative about it.
3. I encourage friends and relatives to go to this company.
4. I recommend this company to people if they want advice on products or services this company could provide.
5. I buy products and services from this company as opposed to from its competitors.